Shafollo Proshad Gupto

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Education

Bangladesh University of Engineering & Technology (BUET) *Bangladesh*

B.Sc. in Naval Architecture and Marine Engineering *Jan 2022 – Jun 2026*

CGPA: 3.77/4.00 • Ranked 4th among 55 students in the department

Relevant Courses: Fluid Mechanics • Marine Hydrodynamics • Computational Fluid Dynamics (CFD) • Numerical Computations Sessional • Resistance and Propulsion of Ships • Basic Thermal Engineering • Heat Transfer • Mechanics of Structure • Finite Element Method (FEM) for Ship Structure

Research Interests

- Computational Fluid Dynamics
- Heat Transfer
- Phase Change Materials

Research Experience

1. Independent collaborative research *2025–2026*

Research Title: ***Numerical Investigation of Melting Enhancement in a Phase Change Material-Filled Enclosure Using K-Shaped Fins: Effects of Fin Angle and Arm Length Ratio***

- Numerically investigated K-shaped aluminium fins for enhancing the melting of low-thermal-conductivity lauric acid PCM in a rectangular cavity using ANSYS Fluent 2024 R1.
- Evaluated five fin-arm-length configurations at fin angles of 10° to 80° and developed the Lower-zone Melting Lag Index (LMLI) to quantify thermal non-uniformity.
- Demonstrated up to 53.35% higher melting efficiency at 40°–50° with an extended lower fin arm and found an inverse relationship between melting efficiency and time-averaged LMLI.

> **My Contributions**
> - Proposed the K-shaped-fin investigation and collaborated on mesh/time-step independence studies and parametric simulations.
> - Analysed and post-processed simulation results, calculated melting efficiency, LMLI and Nusselt numbers, and wrote the original manuscript, including the complete Results and Discussion section.

2. Undergraduate Thesis *Apr 2025 – Jun 2026*

Thesis Title: ***Prediction of Calm-Water Resistance Using Smoothed Particle Hydrodynamics (SPH)***

Supervisor: Dr. Md. Mashiur Rahaman, Professor, BUET

- Investigated SPH for predicting the calm-water resistance of the KCS displacement hull.
- Implemented WCSPH using the GPU-accelerated DualSPHysics framework.
- Developed a numerical towing tank and identified 9 mm as the optimal particle spacing; resistance predictions at $Fr = 0.152$–0.282 showed errors below 6.5%.

> **My Contributions**
> - Developed the DualSPHysics setup and conducted particle-spacing convergence studies to select the optimal resolution.
> - Post-processed and validated resistance predictions against experimental data across five Froude numbers, including resistance-coefficient calculations.

Manuscripts Under Review/Publications

Md Tanvir Anjum Zenith, Adri Das, **Shafollo Proshad Gupto**, Md. Atiqur Rahaman, Md. Khairul Islam, "Numerical Investigation of Melting Enhancement in a Phase Change Material-Filled Enclosure Using K-Shaped Fins: Effects of Fin Angle and Arm Length Ratio." *Case Studies in Thermal Engineering* (Under Review).

Contribution: Conceptualization, Data Curation, Investigation, Software, Writing – Original Draft, Formal analysis.

Engineering Project

Undergraduate Project *Jan 2024 – Feb 2025*

Supervisor: Dr. Nayeb Md. Golam Zakaria

Design of an Inland General Cargo Vessel with 2500 Ton Cargo Capacity

Developed an inland general cargo ship with a 2,500-ton cargo carrying capacity for year-round operation on the Dhaka–Chittagong route. Performed detailed ship design, economic analysis, and assessment of environmental impacts.

Technical Skills

CAD & 3D Modelling	AutoCAD, Rhinoceros 3D
Programming Languages	C++, Fortran, Python
Structural Analysis	ABAQUS
Fluid Flow Analysis	ANSYS FLUENT, DualSPHysics
Ship Design	MAXSURF

Honors and Awards

- **University Dean's List Scholarship** (1st, 2nd and 4th year)
- **University Merit Scholarship** (1st, 2nd and 4th year)
- **3rd Prize** in "Worldwide Ferry Safety Association Design Competition 2026", Team Black Pearl.

Professional Experience

Intern *Jan 2024 – Feb 2025*

Dockyard & Engineering Works Ltd, BANGLADESH NAVY, Narayanganj, Bangladesh

Leadership and Service

- University Class Representative, Department of Naval Architecture and Marine Engineering, BUET *Dec 2022 – Jan 2025*
- Volunteer, MARTEC 2022 – International Conference on Marine Technology *Dec 2022*
- Team Member, Black Pearl, BUET, Worldwide Ferry Safety Association Design Competition *Sept 2025 – Mar 2026*

References

Dr. Nayeb Md. Golam Zakaria
Professor
Dept. of Naval Architecture & Marine Engineering, BUET
Contact:
gzakaria@name.buet.ac.bd
Relationship: Ship design project supervisor.

Dr. Md. Mashiur Rahaman
Professor
Dept. of Naval Architecture & Marine Engineering, BUET
Contact:
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Relationship: Undergraduate thesis supervisor.

Dr. Zobair Ibn Awal
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Dept. of Naval Architecture & Marine Engineering, BUET
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Relationship: WFSA Ferry design competition supervisor.